Week 04 - Tutorial 01

Transcript Language

English

T Hello friends, in this video we will discuss

about basis.

So, first of all what is basis?

Basis is your set, which is collection of linearly independent vector from the vector

space and spanning set of these vectors is the given whole vector space.

Now, consider a vector space v, which is set of all 2 cross 2 matrices over R. So, we know that this is a vector space over R with as usual matrix addition and scalar multiplication

with a matrix.

So, this set form a vector space over R. Now let A be a 2 cross 2 matrix, we are writing

as a11, a12, a21, a22, so this matrix represent a central element in M 2 cross 2 set up all

2 cross 2 matrices.

Now, this matrix we can write it as a11, a10, 01 plus a12, 01, 00 plus a21, 00, 10 plus

a22 00, 01.

So, this matrix is actually linear combination of these matrices.

And we are saying any matrix will be in this form, and this matrix can be generated by

these four matrices.

Now, let us check is these matrices are linearly independent?

So, this is 0 actually here, are linearly independent.

So, let us alpha time 1000 plus beta times 0100 plus gamma 0010 plus delta 0001, which

we can write it is equal to 0.

So, this is the way to checking linearly independent.

Now, we can write this sum as alpha, beta, gamma, delta, this is equal to 0000, this

0 means 0 matrix.

Here, if we compare this imply that alpha is equal to 0, beta is equal to 0, gamma is

equal to 0 and delta is equal to 0 that means, these matrices are linearly independent, and

this matrix can be generated by these four matrices.

And we have seen that these matrices are linearly independent.

It means linear combination of these vectors or we can say matrices, generates whole 2

cross 2 matrices, and these are linearly independent.

So this is form a basis of set of all 2 cross 2 matrices.